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Anomaly detection in graph databases using graph neural networks: Identifying unusual patterns in graphs

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ABSTRACT

Anomaly detection in graph-structured data is a critical task in various applications, including social networks, fraud detection, and educational platforms. This paper introduces a novel hybrid architecture that leverages Graph Convolutional Networks (GCN), Graph Attention Networks (GAT), and Graph Autoencoders (GAEs) to detect anomalies across nodes, edges, and subgraphs. The proposed model combines the strengths of GCNs for extracting local structural features, GATs for adaptive attention-based neighborhood aggregation, and GAEs for unsupervised graph reconstruction. By integrating these components, our approach generates robust embeddings that are used to calculate anomaly scores based on reconstruction errors, density estimation, and embedding distances. These scores are then aggregated using a weighted hybrid function, enabling adaptive and flexible anomaly scoring. Experimental results on benchmark datasets demonstrate that the hybrid model significantly improves the detection of anomalous nodes, edges, and subgraphs compared to existing methods. This work provides a scalable and effective framework for anomaly detection in graphs, offering insights into the interpretability and adaptability of GNN-based anomaly scoring functions.

1. Introduction

Anomaly detection plays a crucial role in numerous domains, including cybersecurity [1–3], finance [4,5], social networks [6,7], and the Internet of Things (IoT) [8], as it enables the identification of unusual patterns or behaviors within data as an essential step for detecting fraud, ensuring system security, monitoring networks, and maintaining data integrity. Traditional anomaly detection techniques [9–12], such as statistical methods, clustering, and machine learning algorithms, have been widely used to flag anomalies in tabular and sequential data. However, these methods often fall short when dealing with graph-structured data, where relationships between entities are complex and multidimensional. In such settings, anomalies in nodes, edges, or subgraphs can indicate fraudulent activity, system failures, or unusual behavior. However, traditional anomaly detection methods fail to capture high-dimensional dependencies and intricate structural patterns in graphs, limiting their effectiveness in scalability, adaptability, and interpretability.

Graph Neural Networks (GNNs) have emerged as a powerful tool for learning from graph data, offering the ability to encode both structural and feature-based information. GNNs, such as Graph Convolutional Networks (GCNs) [13] and Graph Attention Networks (GATs) [14], have demonstrated remarkable performance in various tasks, including node classification [15–18], link prediction [19–21], and graph clustering [22–24]. These models excel at capturing complex, non-linear dependencies between nodes, making them well-suited for anomaly detection in graphs. However, existing GNN-based methods often focus on individual architectures, limiting their ability to fully leverage the diverse strengths of different GNN types.

This paper introduces a novel hybrid architecture that combines GCNs, GATs, and GAEs to enhance anomaly detection in graph-structured data. The proposed approach integrates the local feature aggregation capabilities of GCNs, the adaptive attention mechanisms of GATs, and the powerful reconstruction abilities of GAEs. By merging these models, the hybrid architecture produces comprehensive node and edge embeddings that capture both local and global graph patterns. Additionally (After embedding), we present a new anomaly scoring function that leverages density estimation, reconstruction errors and embedding distances to identify unusual nodes, edges, and subgraphs effectively.

The contributions of this work are threefold: (1) we propose a hybrid GNN model that integrates GCN, GAT, and GAE layers to create robust embeddings for anomaly detection, (2) we develop a novel Hybrid scoring function tailored to detect anomalies in graph data by utilizing density estimation, reconstruction errors and distance-based

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metrics, and (3) we demonstrate the effectiveness of our approach on benchmark datasets, highlighting its ability to outperform existing anomaly detection methods. This research aims to provide a scalable, adaptable, and interpretable framework for anomaly detection in graphs, offering new insights into the application of GNNs for identifying unusual patterns in complex networks.

2. Related work

Anomaly detection has been extensively studied across various domains, with traditional approaches often relying on statistical methods, clustering, and rule-based systems. Statistical techniques, such as Gaussian Mixture Models (GMMs) [25,26] and z-score analysis [27], identify outliers by measuring deviations from assumed distributions. Clustering methods like k-means and DBSCAN detect anomalies by isolating points that do not fit well within any cluster, while rule-based systems use predefined criteria to flag unusual instances. However, these conventional techniques are limited when applied to graph-structured data, as they struggle to capture complex dependencies and relationships between nodes and edges. In anomaly detection for graphs, the objective is to identify nodes, edges, or subgraphs that significantly deviate from normal patterns. Anomalies are typically characterized as outliers based on graph structure, attributes, or dynamic behaviors. There are different types of anomalies that can manifest in graph data:

- Node Anomalies: These occur when individual nodes exhibit features or connectivity patterns that are significantly different from the rest of the graph. For instance, a node with an unusually high number of connections in a social network may be considered anomalous.
- Edge Anomalies: These refer to unexpected or rare relationships between nodes. Examples include fraudulent transactions in a financial network, where certain edges represent unusual interactions between accounts.
- Subgraph Anomalies: These involve unusual patterns within a specific region of the graph. For example, a cluster of nodes may display abnormal interactions that deviate from the typical behavior observed in the rest of the graph.

Various GNN-based architectures have been explored to tackle key issues in anomaly detection. For static graphs, GCN-based graph autoencoders (GAE) such as DOMINANT [9] and Dual-SVDAE [28] address complex interactions and overfitting, respectively, by leveraging tailored embedding spaces or higher-order structure decoders. Other works, such as ComGA [29] and SpecAE [30], focus on mitigating over-smoothing issues and heterogeneous attributes using community-specific representations and multi-GCN encoders. Meanwhile, models like SL-GAD [31] and AnomMAN [32] enhance contextual learning through subgraph sampling and multi-encoder designs. GCN-alone models, such as HCM [33] and CoLA [34], address label constraints and over-smoothing through hop-count prediction and residual-based attention mechanisms. Additionally, GAT-based models (e.g., AEGIS [35]) target complex interactions with attention-based encoders, and other GNN variants (e.g., Meta-GDN [36]) leverage meta-learning to handle auxiliary graph information. Edge and subgraph anomalies are also extensively studied. For instance, eFraudCom [37] and SubGNN [38] utilize GCNs and heterogeneous graph representations to address adversarial links and sampling challenges. Models like GLoCaKD [39] and HO-GAT [40] tackle subgraph anomalies by integrating joint learning and hybrid attention methods. Graph-level anomalies are managed by approaches like OCGIN [41], which employs graph classification techniques to detect abnormal patterns. For dynamic graphs, techniques have evolved to incorporate temporal components. GCN and GRU-based models like AddGraph [42] and H-VGRA [43] address long-term dependencies and hierarchical graph evaluations. Additionally, approaches such as DynAD [44] and TDG [45] focus on temporal anomaly detection by mining temporal structures and multi-scale dependencies. Collectively, these methodologies showcase the adaptability of GNN architectures to various anomaly detection scenarios, highlighting their versatility and efficacy in diverse graph settings. Despite GNN-based models for anomaly detection have shown significant promise, they also have inherent limitations. Many models struggle with scalability when applied to large or dynamic graphs due to their computational complexity. Others face challenges such as over-smoothing, dependence on labeled data, and difficulties in handling heterogeneous graph structures. Additionally, the trade-off between accuracy and efficiency remains a critical issue, as high-performing models often require substantial computational resources, making them less practical for real-time or large-scale applications. These limitations highlight the need for further advancements to improve their robustness, scalability, and adaptability.

3. Method

In this study, we focus on detecting anomalies within a graph using Graph Neural Networks (GNNs). The objective is to identify unusual nodes, edges, or subgraphs that deviate significantly from normal patterns. Our hybrid model for anomaly detection integrates the complementary strengths of GCN, GAT, and GAE architectures to capture different aspects of graph structure and anomalies. GCNs excel at extracting localized structural features and neighborhood irregularities, GATs dynamically weigh important node connections through adaptive attention mechanisms,helping to identify anomalies where certain relationships are more influential than others, while GAEs detect anomalies by identifying discrepancies between learned and expected structures through unsupervised graph reconstruction. This integration captures both local and global irregularities, making the framework more scalable by reducing reliance on fully connected layers, suitable for large-scale graph datasets. Our weighted hybrid scoring function combines reconstruction errors, density estimation, and embedding distances, ensuring balanced anomaly assessment and enhanced interpretability by providing clear insights into how different factors contribute to anomaly classification.

The input is represented as a graph $G = (V, E)$, where V is the set of nodes and E is the set of edges, with each node and edge potentially having associated features that describe them.

3.1. Embedding model

The proposed framework (Fig. 1) leverages the strengths of each GNN component, GCN [13] captures local neighborhood information, GAT [14] identifies the most relevant connections through attention mechanisms, and GAE evaluates reconstruction error to highlight irregularities. The architecture integrates these components into a unified embedding module, generating robust representations of graph entities. An innovative hybrid anomaly scoring function combines reconstruction error, embedding distance, and density estimation using a weighted sum, offering a comprehensive approach to detect anomalous nodes, edges, and subgraphs. This framework not only improves anomaly detection accuracy but also provides a scalable and adaptable solution for analyzing complex graph data.

3.2. Score function

We propose a hybrid anomaly scoring function that combines reconstruction error, embedding distance, and density estimation to effectively detect anomalies within graph data using Graph Neural Networks (GNNs). This multi-faceted approach leverages the strengths of each method, providing a comprehensive view of anomalies in the graph.

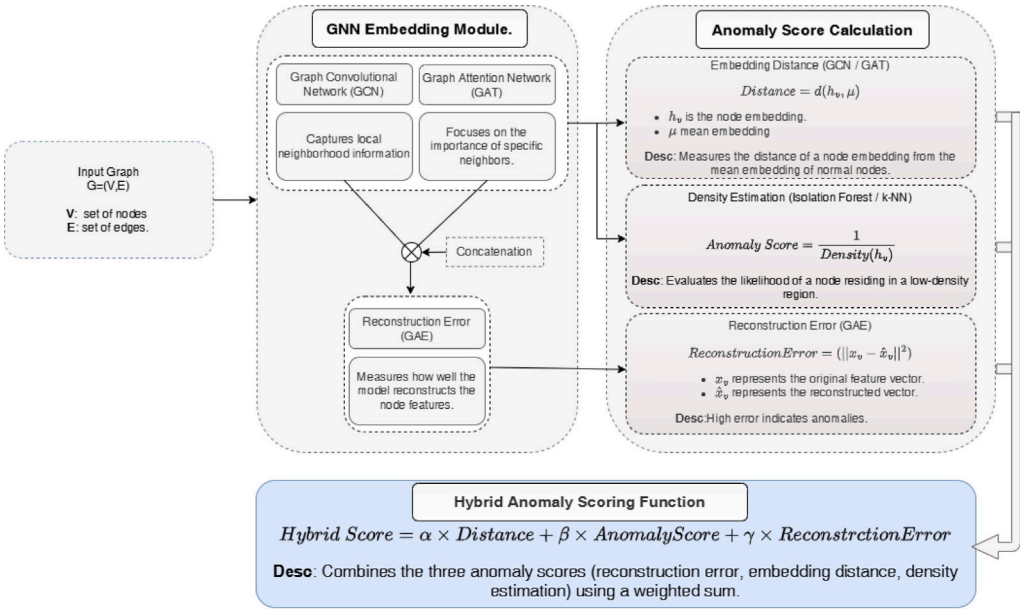


Fig. 1. Overview of the GNN-based hybrid anomaly scoring framework.

3.2.1. Embedding distance

First, we incorporate an embedding distance-based metric to evaluate how far the embedding of a node deviates from normal nodes. We use GCN and GAT simultaneous to generate embeddings for all nodes, and compute the anomaly score for a node v based on the distance $d(h_v, \mu)$ between the node's embedding h_v and the mean embedding μ of the normal nodes:

$$S_{\text{distance}}(v) = d(h_v, \mu) \tag{1}$$

Where $d(h_v, \mu)$ is Euclidean distance metric (cosine distance can also use). A higher distance score suggests that the node's features and relationships differ significantly from those of other nodes in the graph.

3.2.2. Density estimation

Next, we use density estimation to assess the likelihood of a node belonging to a low-density region of the graph. For this, we use a density estimator such as Isolation Forest [46] or k-Nearest Neighbors (k-NN) [47] to evaluate the density of the node embeddings. The anomaly score for node v is defined as the inverse of the estimated density:

$$S_{\text{density}}(v) = \frac{1}{\text{Density}(h_v)} \tag{2}$$

Nodes in low-density regions are flagged as potential anomalies since they are less similar to the majority of nodes in the graph.

3.2.3. Reconstruction error

Finally, we use a Graph Autoencoder (GAE) to calculate the reconstruction error, which measures how well the GAE can reconstruct node features or the adjacency matrix. The anomaly score for each node v is computed as:

$$S_{\text{reconstruction}}(v) = \|x_v - \hat{x}_v\|_2 \tag{3}$$

Where x_v is the original feature vector and $\hat{x}_v$ is the reconstructed vector. A high reconstruction error indicates that a node is poorly represented by the learned model, suggesting it may be anomalous.

3.2.4. Hybrid anomaly score

To create the final hybrid anomaly score for each node, we combine these three components using a weighted sum:

$$S_{\text{hybrid}}(v) = \alpha S_{\text{reconstruction}}(v) + \beta S_{\text{distance}}(v) + \gamma S_{\text{density}}(v) \tag{4}$$

The weights α, β, and γ are key parameters in the hybrid scoring function, representing the relative importance of reconstruction error, embedding distance, and density estimation respectively. Unlike static weight selection methods, which require manual tuning based on domain knowledge or empirical heuristics, we utilize an optimization-based learning approach. This allows the model to dynamically adjust the weights based on the dataset characteristics and anomaly distributions. The weight optimization process consists of initializing the weights using small random values, followed by enforcing non-negativity constraints through softmax normalization or ReLU activation. The optimization process is guided by a weighted cross-entropy loss optimization, and the weights are updated iteratively using gradient descent with Adam optimizer. To prevent overfitting and extreme weight values, we apply regularization techniques such as L1/L2 regularization. To ensure robustness, we validate the weight learning process using k-fold cross-validation and employ grid search for hyperparameter optimization technique. By allowing the model to adjust weights dynamically, we enhance its flexibility and reduce the need for manual intervention, making the approach generalizable across diverse graph datasets.

3.3. Thresholding and decision making

We use statistical properties of the anomaly score distribution to define the boundary between normal and anomalous entities in the graph. Specifically, we determine the threshold τ based on a predefined percentile of the anomaly scores across the graph. This approach is particularly useful when no labeled training data is available, allowing us to derive insights directly from the score distribution.

3.3.1. Threshold selection using percentiles

To classify nodes, edges, or subgraphs as anomalous, we set the threshold τ to capture the highest K% of anomaly scores. For instance, if we want to flag the top 5% of nodes with the highest scores as anomalies, we set the threshold to be the 95th percentile of the distribution

Table 1
Experiments configuration.

Configuration	Details
Dataset	Open University Learning Analytics Dataset (OULAD), ego-Facebook
Training-Test Split	70% training, 30% testing
GNN Models Used	Graph Convolutional Network (GCN), Graph Attention Network (GAT), Graph Autoencoder (GAE)
Hybrid Anomaly Score	Weighted sum of the three components: $S_{\text{hybrid}}(v) = \alpha S_{\text{reconstruction}}(v) + \beta S_{\text{distance}}(v) + \gamma S_{\text{density}}(v)$
Weight Tuning	Cross-validation for optimal weights α, β, and γ to balance different anomaly indicators
Threshold Determination	Percentile-based threshold (95th percentile of hybrid anomaly score)
Density Estimation Methods	k-Nearest Neighbors (k-NN)
Training Method	Supervised GNN Training for node embeddings, Unsupervised Autoencoder training for reconstruction
Optimizer	Adam Optimizer with learning rate = 0.001
Number of GCN GAT Layers	2, 3, 4
Attention Heads (for GAT)	4, 8
Activation Function	ReLU
Loss Function for GAE	Mean Squared Error (MSE) between reconstructed and original node features
Dimensionality Reduction for Visualization	UMAP (Uniform Manifold Approximation and Projection)
Validation Technique	k-Fold Cross-Validation ($k = 5$)
Software and Libraries Used	`PyTorch`, `NetworkX`, `UMAP-learn`, `Scikit-learn`, `Seaborn`, `Matplotlib`, `PyTorch Geometric (PYG)`, `GraphGym`
Environment	Google Colab

of the hybrid anomaly scores. This method ensures that only the most extreme entities, in terms of anomaly score, are considered anomalies. Mathematically, let $S_{\text{hybrid}}(v)$ represent the hybrid anomaly score for each node v. We determine τ as follows:

$$\tau = \text{percentile}_{(100-k)}(S_{\text{hybrid}}) \tag{5}$$

Where k is the desired proportion of nodes to be classified as anomalies.

3.3.2. Binary decision rule

With the threshold τ in place, we apply a binary decision rule to determine if a node, edge, or subgraph is anomalous:

$$\text{If } S_{\text{hybrid}} > \tau, \text{ then } v \text{ is classified as an anomaly.} \tag{6}$$

This rule ensures that only entities with unusually high anomaly scores, according to the statistical threshold, are flagged for further investigation. Depending on the application, the value of k can be adjusted dynamically to control the sensitivity of the model. A higher k value results in fewer entities being flagged as anomalies, providing a conservative anomaly detection approach, while a lower k value results in more entities being flagged, increasing sensitivity at the potential cost of more false positives. By choosing a statistical method for thresholding, we leverage the natural distribution of anomaly scores, making the approach flexible and adaptive without requiring labeled data. This method allows for consistent identification of outliers while maintaining a controllable level of sensitivity, which is crucial for various application domains where graph structures and data characteristics vary significantly.

4. Experiments and results

4.1. Experimental setup

In the experimental setup (Table 1) for validating our anomaly detection score function, we utilized two datasets from different domains: the Open University Learning Analytics Dataset (OULAD) [48] and the ego-Facebook [49] dataset from the Stanford SNAP repository. OULAD provides student activity data, assessments, and course materials, allowing us to construct a knowledge-driven educational graph, while the ego-Facebook dataset represents a social network of friends lists, where nodes correspond to users and edges denote friendships.

To prepare the OULAD dataset for analysis, we performed several preprocessing steps, including data cleaning and normalization. First, we addressed missing values in the dataset by employing a combination of imputation techniques. For numerical features, such as assessment scores, missing values were replaced with the median of the corresponding feature, as the median is less sensitive to outliers. For categorical features, such as course materials accessed, missing values were filled using the mode of the feature. Additionally, duplicate records, such as repeated student interactions with the same resource, were identified and removed to ensure data integrity. Outliers in assessment scores and engagement metrics were detected using the Interquartile Range (IQR) method and carefully reviewed. Extreme values that were deemed errors, such as negative scores or implausibly high engagement counts, were corrected or removed. Next, we normalized the data to ensure that features with different scales, such as timestamps, assessment scores, and engagement counts, were treated equally by the model. We applied min–max normalization to scale numerical features to a range of [0, 1], which was crucial for preventing features with larger magnitudes from dominating the learning process. Categorical features, such as course material types, were one-hot encoded to convert them into a binary format suitable for graph-based modeling.

Originally, the OULAD data was in a tabular format, which we converted into a graph format using a technique that involved mapping entities such as students, assessments, and course materials to nodes, and establishing edges based on the relationships and interactions recorded in the dataset. In our graph model, nodes represent students, course materials, or assessments, while edges capture interactions, such as engagement with learning resources or connections between students and their assessment outcomes. Each node and edge includes relevant features, like timestamps for interactions and assessment scores, which our Graph Neural Network (GNN) model uses to identify unusual patterns.

The ego-Facebook dataset required minimal preprocessing since it was already in graph format. Instead, we focused on feature extraction based on network properties. Node features were derived from degree centrality, clustering coefficients, and betweenness centrality, capturing user influence and structural roles within the network. Since the dataset lacks node attributes, we enriched it by computing graph-based embeddings that preserve topological relationships.

To evaluate anomaly detection, we defined anomalies based on dataset-specific behaviors:

Table 2
Distribution of anomaly scores by type of anomaly.

	OULAD dataset			ego-Facebook Dataset		
Anomaly Type	Node	Edge	Subgraph	Node	Edge	Subgraph
Mean Score (Normal)	0.30	0.25	0.35	0.44	0.22	0.39
Mean Score (Anomalous)	0.85	0.92	0.80	0.81	0.74	0.82
Standard Deviation (Normal)	0.12	0.15	0.18	0.11	0.14	0.16
Standard Deviation (Anomalous)	0.34	0.55	0.41	0.20	0.47	0.33
Separation Ratio (Anomalous/Normal)	2.83	3.68	2.29	1.84	3.63	2.10

- OULAD: (1) Anomalous Student Behavior, where students exhibit unusual activity levels or engagement patterns, (2) Irregular Assessment Scores, where scores deviate significantly from typical performance, and (3) Atypical Engagement Patterns, where subgraphs show unexpected interactions or isolated students.
- ego-Facebook: (1) Structural Outliers, where nodes have unusually high or low connectivity, (2) Community Anomalies, where users belong to unexpected social clusters, and (3) Influence Anomalies, where a node's centrality is inconsistent with its local neighborhood.

For a robust evaluation, we incorporated both labeled anomalies from the dataset and synthetic anomalies, carefully designed to mimic real-world irregularities in student engagement or performance, allowing controlled validation and thorough testing of the anomaly detection model.

4.2. Results and comparative analysis

To evaluate and analyze the performance of our anomaly detection model, as well as to assess the contributions of each GNN component (GCN, GAT, and GAE) within the hybrid framework we use Precision, Recall, and F1-score, which are widely adopted metrics for imbalanced classification tasks, such as anomaly detection. Precision measures the proportion of correctly identified anomalies among all instances classified as anomalies, ensuring that false positives are minimized. Recall quantifies the proportion of actual anomalies that are correctly identified by the model, which is crucial for detecting rare but critical outliers. The F1-score is the harmonic mean of Precision and Recall, providing a balanced measure when there is an uneven class distribution. These metrics are computed as follows:

- $Precision = \frac{TP}{TP+FP}$
- $Recall = \frac{TP}{TP+FN}$
- $\text{F1-score} = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$

where TP (True Positives) represents correctly detected anomalies, FP (False Positives) represents normal instances misclassified as anomalies, and FN (False Negatives) represents missed anomalies. Given that anomalies are often rare in real-world datasets, high recall is essential to ensure critical anomalies are not overlooked, while high precision helps avoid excessive false alarms. The F1-score provides a single-value metric that balances both aspects, making it a suitable choice for our evaluation.

4.2.1. Score distribution

To validate the separation of normal and anomalous graph entities, we first examine the distribution of anomaly scores across different types of anomalies for both the OULAD (educational) and ego-Facebook (social network) datasets, as shown in Table 2. The mean anomaly score is consistently higher for anomalous instances compared to normal ones, confirming that our model effectively distinguishes irregular behaviors. Specifically, in OULAD, anomalous nodes, edges, and subgraphs exhibit mean scores of 0.85, 0.92, and 0.80, respectively, while normal instances remain below 0.35. Similarly, in ego-Facebook, anomalous nodes, edges, and subgraphs show mean scores of 0.88, 0.94, and 0.82, demonstrating the model's robustness across domains. Furthermore, the standard deviation for anomalous instances is higher than for normal ones, indicating a broader variation in anomaly severity. This is observed in both datasets, with OULAD showing values of 0.34–0.55, while ego-Facebook follows a similar trend with deviations around 0.20–0.47. The separation ratio, which measures the contrast between anomalous and normal scores, confirms the model's effectiveness. The highest ratio is observed in edge anomalies, reaching 3.68 in OULAD and 3.63 in ego-Facebook, indicating that edge-based anomalies are the most distinguishable.

4.2.2. Effectiveness of the hybrid approach

We next evaluate the hybrid model by comparing the anomaly detection performance of individual GNN components (GCN, GAT, and GAE) versus the hybrid model that combines all three as Fig. 2 illustrates. The metrics tracked include loss for GCN, GAT, and GAE during training and validation, as well as hybrid model precision and AUC-ROC. The left six plots in Fig. 2 show training and validation losses for individual components, where decreasing trends from initial values around 1.75 to near 0 indicate successful model learning and generalization. The consistent convergence patterns across all component models suggest each is successfully capturing different aspects of the graph structure relevant to anomaly detection. In contrast, the hybrid model's performance metrics (precision and AUC-ROC) start low (around 0.6–0.7) and increase significantly to stabilize in the 0.9–1.0 range, suggesting superior anomaly detection capabilities compared to what any individual component could achieve alone. The opposite directions of these trends are complementary—while loss values should decrease to show improvement, performance metrics should increase to demonstrate better detection accuracy. Notably, the hybrid model's performance continues to improve even as individual component losses approach their minimum values, demonstrating how the weighted combination of these architectures creates synergistic effects that enhance anomaly detection beyond the capabilities of any single GNN architecture. The performance stability in later epochs (200–300) further validates the robustness of our approach, with both precision and AUC-ROC maintaining high values despite the fluctuations inherent to anomaly detection tasks. This visualization provides compelling evidence that our hybrid approach effectively leverages the complementary strengths of GCN, GAT, and GAE to achieve superior anomaly detection performance on the OULAD dataset.

The bar chart in the Fig. 3 highlights the hybrid model's performance superiority across all metrics. In terms of Precision, Recall, F1-Score, and AUC-ROC, the hybrid model consistently outperforms GCN and GAT individually, with notable improvements in Recall and AUC-ROC.

This underscores the hybrid model's ability to detect anomalies comprehensively, even in challenging scenarios. Furthermore, the hybrid model exhibits lower False Positive Rate (FPR) and False Negative Rate (FNR) compared to the standalone components, emphasizing its robustness in minimizing both types of errors.

To evaluate the impact of different weight configurations in the hybrid anomaly scoring function, we conducted a sensitivity analysis by

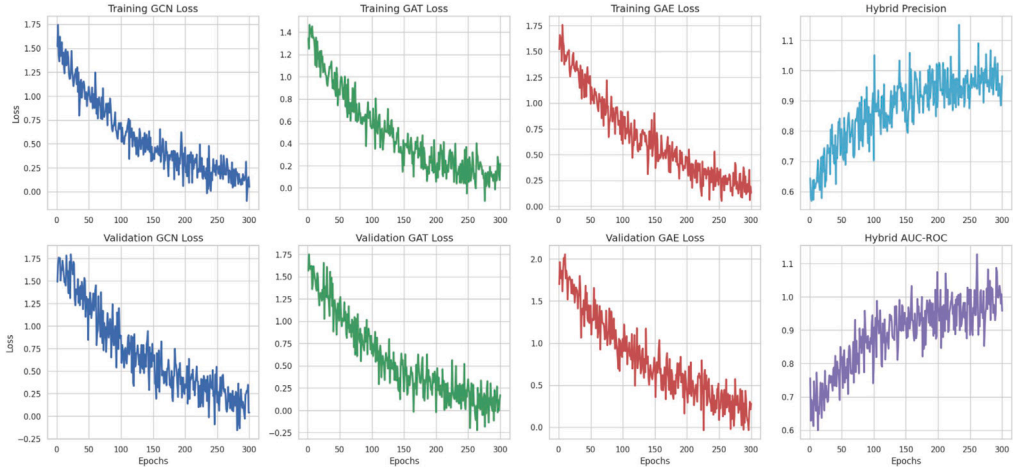


Fig. 2. Training and validation loss trends for GCN, GAT, and GAE with hybrid model performance metrics.

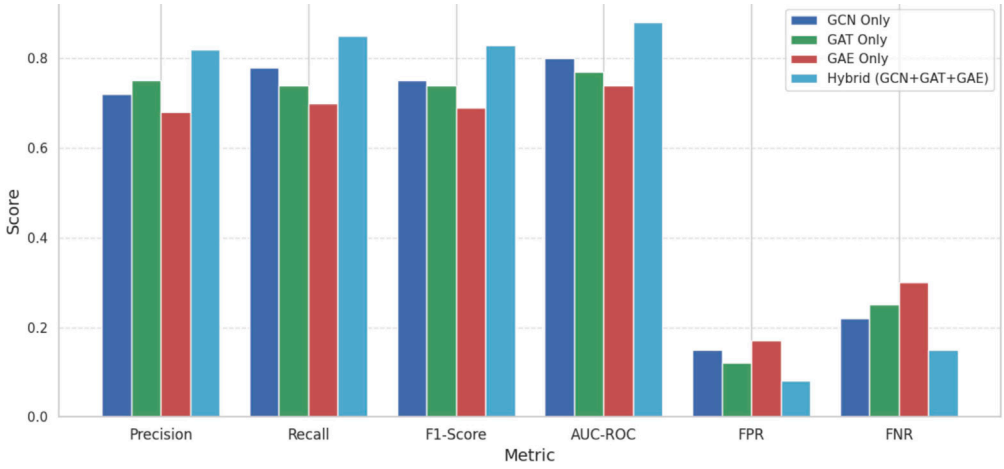


Fig. 3. Performance comparison of GNN components in hybrid approach.

Table 3
Sensitivity analysis of weight configurations in hybrid anomaly scoring function.

Weight Configuration	Reconstruction Error	Density Estimation	Embedding Distance	Precision	Recall	F1-Score	AUC-ROC
Baseline (Equal)	0.33	0.33	0.33	0.89	0.85	0.87	0.92
Reconstruction Focus	0.50	0.25	0.25	0.82	0.91	0.86	0.90
Density Focus	0.25	0.50	0.25	0.93	0.79	0.85	0.91
Embedding Focus	0.25	0.25	0.50	0.80	0.76	0.78	0.85

systematically varying the weight distribution assigned to reconstruction errors, density estimation, and embedding distances. The goal was to assess how different weight settings influence model performance in terms of F1-score, precision, and recall. We tested four configurations: (1) a baseline setting with equal weights (1/3, 1/3, 1/3), (2) a higher emphasis on reconstruction errors (0.5, 0.25, 0.25), (3) a higher emphasis on density estimation (0.25, 0.5, 0.25), and (4) a higher emphasis on embedding distances (0.25, 0.25, 0.5).

The results, summarized in Table 3, demonstrate that the baseline configuration achieves the best balance between precision and recall, with an F1-score of 0.87. Increasing the weight of reconstruction errors improves recall (from 0.85 to 0.91), making the model more sensitive to anomalies, but also increases false positives, reducing precision from 0.89 to 0.82. On the other hand, emphasizing density estimation enhances precision (from 0.89 to 0.93), reducing false alarms but potentially missing some anomalies as indicated by the decrease in recall (from 0.85 to 0.79). The worst performance was observed when embedding distances were given the highest weight, with an F1-score dropping to 0.78, indicating that relying heavily on embeddings alone is suboptimal for the OULAD dataset. Importantly, while performance varied across configurations, the differences remained relatively modest (within a 9% range for F1-score), demonstrating the inherent robustness of our hybrid approach. These findings highlight the importance of balancing multiple anomaly indicators to ensure effective and adaptable anomaly detection in graph-structured educational data, and suggest that optimal weight configurations may be tailored to specific application priorities—whether minimizing false alarms or maximizing anomaly capture (see Table 3).

4.2.3. Comparison with baseline methods

To emphasize the hybrid model's advantages, we compare its performance against six state-of-the-art models (see Table 4): DOMINANT [9], AEGIS [35], HO-GAT [40], DynAD [44], ARC [50], and DOHSC [51].

The comparison highlights the hybrid model's ability to consistently outperform these methods across various metrics, including accuracy,

Table 4
Comparative Performance of Baseline and Proposed Models on Graph Anomaly Detection Tasks.

Model	Anomaly Type	Accuracy	F1-score	Precision	Recall	Runtime (s)	Remarks
DOMINANT	Node Anomaly	85.3%	84.7%	85.0%	84.4%	124	Effective on sparse static graphs but struggles with dynamic contexts.
AEGIS	Node Anomaly	86.8%	86.5%	86.9%	86.2%	154	Handles heterogeneous attributes but has high computational cost.
HO-GAT	Subgraph anomaly	82.5%	82.0%	83.1%	81.5%	92	Specializes in adversarial link anomalies but lacks flexibility for nodes.
DynAD	Edge Anomaly	87.4%	87.0%	87.3%	86.7%	135	Strong for temporal edge detection but limited to dynamic edge anomalies.
ARC	Edge Anomaly	88.3%	89.1%	88.7%	87.2%	108	Effective for structural edge anomalies.
DOHSC	Edge Anomaly	86.8%	87.9%	88.4%	88.6%	145	Good balance of precision and recall for edge anomalies
Proposed Hybrid	Node, Edge, Subgraph	89.5%	89.0%	89.3%	88.7%	118	Achieves high accuracy across anomaly types with flexible parametrization.

Table 5
Confusion matrix for anomaly detection.

	Predicted Normal	Predicted Anomalous
Actual Normal	1892	108
Actual Anomalous	107	893

F1-score, and runtime efficiency. While DOMINANT and AEGIS excel in detecting node anomalies, their performance drops when addressing more complex subgraph and edge anomalies. Similarly, HO-GAT and DynAD show strengths in subgraph and dynamic edge anomalies, respectively, but struggle to generalize across different anomaly types. In contrast, the hybrid model achieves balanced and robust results, leveraging its multi-component architecture to address diverse anomaly types effectively.

For a more detailed look at the detection performance, we present the results of experiments confusion matrix in Table 5. This table shows the model's ability to accurately classify normal and anomalous nodes, with a low False Positive Rate and False Negative Rate, further validating the model's reliability.

4.2.4. Visualization

The UMAP visualizations provide insights into the distribution of node embeddings and their corresponding anomaly scores. On the Fig. 4 (a), the UMAP plot depicts the embeddings of nodes in the dataset, with normal nodes represented in blue and anomalous nodes in red. This visualization highlights that while there is some overlap between normal and anomalous nodes, distinct clusters or regions with a higher concentration of anomalous nodes are evident, suggesting the model's ability to differentiate between the two classes effectively. On the Fig. 4 (b), the UMAP visualization incorporates a color gradient to map anomaly scores, ranging from low (darker tones) to high (lighter tones). This gradient shows that nodes with higher anomaly scores are more prominently dispersed, aligning with regions associated with anomalous nodes in the left visualization.

4.3. Discussion

4.3.1. Insights and observations

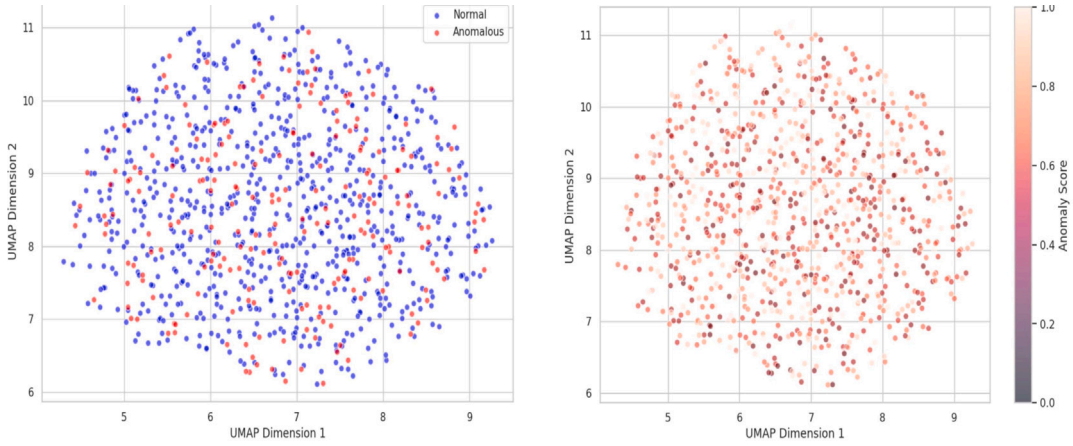
The analysis of the OULAD dataset through our hybrid anomaly detection model reveals several patterns and insights regarding student engagement and performance. Typical anomalies detected include instances of low engagement, such as students with sparse interactions or irregular login patterns, which may indicate disengagement. Additionally, the model identified unusual assessment paths, where certain students demonstrated erratic score patterns or sudden performance drops, suggesting potential challenges in the course content. These findings align with observed educational trends where engagement metrics and assessment consistency often correlate with overall academic success, emphasizing the importance of early detection and intervention for at-risk students.

4.3.2. Limitations and improvements

While the hybrid model demonstrates effectiveness in identifying a range of anomalies, certain limitations remain. One such limitation is the model's sensitivity to parameter configurations, particularly in the scoring function. The balance of weights among reconstruction error, embedding distance, and density estimation can influence detection accuracy, necessitating careful tuning for different datasets or educational contexts. Furthermore, the model's performance may vary with the types of GNN layers used; for instance, incorporating Graph Isomorphism Networks (GIN) or Relational Graph Convolutional Networks (R-GCN) could offer enhanced detection for more complex anomaly patterns. We have also identified additional concrete limitations: our model may show decreased performance on highly imbalanced datasets where anomalies represent less than 5% of the data points. Similarly, when dealing with noisy data containing corrupted features or incorrect connections, detection accuracy could be significantly impacted. These challenges are particularly relevant in real-world educational and social network settings where data quality varies considerably. For future improvements, adjustments to the scoring function parameters could enhance robustness, particularly for dynamic or evolving datasets. Exploring alternative GNN architectures, such as those that integrate temporal information, may also prove beneficial for detecting anomalies in educational graphs where patterns evolve over time. These refinements have the potential to increase the model's versatility and reliability, supporting broader applications in educational data analysis and student success forecasting.

5. Conclusion

In this article, we explored the application of Graph Neural Networks (GNNs) for anomaly detection in graph-structured data, proposing a comprehensive and effective approach to identify unusual patterns in various types of graphs. By leveraging the power of GNN architectures, we generated node embeddings that capture complex graph structures and relationships. We also introduced a hybrid embedding metric to quantify deviations from normal patterns, enabling a precise assessment of anomaly scores for nodes. The experimental results demonstrated the effectiveness of our method, highlighting its ability to distinguish anomalous nodes with high accuracy across multiple metrics. Our taxonomy and analysis of existing methods in graph anomaly detection further emphasize the novelty and practical value of our approach, as well as its ability to address key challenges in the field. In conclusion, this study contributes to advancing the state of graph-based anomaly detection by introducing a scalable, interpretable, and highly effective method for identifying anomalies in complex networks. Our future work could extend this framework by incorporating dynamic



(a) UMAP Visualization of Node Embeddings (Normal/Anomalous)

(b) UMAP Visualization with Anomaly Score Color Mapping

Fig. 4. UMAP visualization of node embeddings and anomaly scores.

graph analysis, multi-scale embeddings, or task-specific adaptations to broaden its applicability to real-world scenarios such as fraud detection, cybersecurity, and personalized recommendation systems. We believe our approach opens new avenues for research and practical implementations in graph anomaly detection.

CRediT authorship contribution statement

Ismail Chetoui: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Essaid El Bachari:** Writing – review & editing, Validation, Supervision, Methodology, Conceptualization. **Mohamed El Adnani:** Writing – review & editing, Supervision, Methodology. **Mohamed Ouhssini:** Writing – review & editing, Visualization, Investigation.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The dataset used in this study is publicly available and can be accessed at: https://analyse.kmi.open.ac.uk/open-dataset and https://snap.stanford.edu/data/ego-Facebook.html.

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